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(71) Applicant (for all designated States except US): **ADVANCED MATERIALS-JTJ S.R.O.** [CZ/CZ]; Kamenné ehrovice 23, 273 01 Kamenné ehrovice (CZ).

(72) Inventors; and

(75) Inventors/Applicants (for US only): **PROCHÁZKA JR., Jan** [CZ/CZ]; Kamenné ehrovice 23, 273 01 Kamenné ehrovice (CZ). **PROCHÁZKA SR., Jan** [CZ/CZ]; Kamenné ehrovice 23, 273 01 Kamenné ehrovice (CZ).

(74) Agent: **KUBICKOVÁ, Kvetoslava**; Czech Technical University in Prague, the Business and Innovation Centre

of the Czech Technical University, Zikova 4, 166 33 Praha 6 (CZ).

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(54) Title: TITANIUM DIOXIDE CATALYST STRUCTURE FOR PROCESSES UP TO 1000 °C AND THE MANUFACTURING THEREOF

(57) Abstract: The TiO₂ catalyst structure consisting of TiO₂ nano-particles in the anatase crystal form, doped with 0.05 - 5 wt% phosphorus on the TiO₂ basis, organized in the circular planar aggregates with the specific surface area ranging from 40 to 120 m²/g, suitable for catalytic processes at the temperature up to 800°C, and the TiO₂ catalyst structure of with the morphology of the aggregated compact particles, with the specific surface area from 20 to 40m²/g, suitable for the catalytic processes at the temperature up to 1000°C. Active substances selected from the group consisting of silver, copper, gold, platinum metals, nickel, molybdenum and metal oxides except for alkaline metals oxides can be applied onto the surface of both types of the structure.

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